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Assignment 4

[redacted] Data Science

Que 1 ⇒ Explain the concepts of data wrangling and its importance in data analysis.

Ans ⇒ Data Wrangling involves cleaning, transforming, and structuring raw data into a usable format for analysis.

Key concepts:-

It includes steps like gathering data from the sources, assessing quality issues, cleaning errors or duplicates, transforming formats (e.g., merging datasets) and validating results. Also called data munging it handles messy, inconsistent data from databases files or APIs.

Importance:-

Data Wrangling ensures accurate, reliable analysis by removing biases and errors, boosting model performance in machine learning, and saving time for insights over cleanup, Without it, raw data stays unusable leading to flawed decisions.

Que ⇒ Explain string manipulation in Python with suitable code examples?

Ans ⇒ String manipulation in python involves operations like slicing, concatenating, and formatting strings, using built in methods.

Core Operations:-

Strings are immutable sequences; access characters via indexing (e.g., s[0] or slicing (s[1:4] for substring). Concatenate with '+' ("Hello" + "World" → "HelloWorld") or * for repetition. ("hi" * 3 → "HiHiHi").

common methods:-

Use upper()/lower() for case changes ("hello".upper() → "HELLO"), replace() for substitutions ("hello".replace("l","p") → "heppop", and split() for lists → ("a, b, c".split(",") → ["a", "b", "c"]) formatting via f-strings: name = "Abhinav"; f"Hello, {name}!" → "Hello, Abhinav".

Example -

```
text = "Python is ~~full~~ fun!"
Clean = text.strip().lower().replace("fun", "great") #"python is great!"
Print(clean.split()) #['python', 'is', 'great!']
```

Que 3 -> Explain the general techniques and programming practices used for efficient handling of large datasets in data science.

Ans -> Efficient Handling of large datasets in datascience uses - Chunking vectorisation and distributed tools to manage - memory & speed.

Key techniques:

Process data in chunks (e.g, Pandas chunksize) or use libraries like - Dask / Vaex for lazy/out of core computation on datasets to big for - RAM. Opt for efficient formats (Parquet, HDF5), with compression - Sampling and early filtering to shrink data volume.

Programming Practices:-

Leverage vectorised operations (Numpy / Pandas) over loops, optimize dtypes (e.g. int8, category) and ~~practice~~ parallelize with Spark / Hadoop for distri - buted processing. Stream data for realtime needs and profile memory with tools like (memory_profiler).

Que 4 => Explain ~~the~~ different data transformation techniques used in preprocessing.

Ans => Data transformation techniques in preprocessing convert raw - inconsistent data into a clean, standardized format suitable for analysis and modeling, addressing issues like varying - scales, noise and incomplete types.

(P.T.O) →

Normalisation & Standardization:- Normalization scales features to a fixed range, typically using min-max scaling: $x' = \frac{x - \min(x)}{\max(x) - \min(x)}$. This prevents feature with larger ranges from dominating distance ~~between~~ based algorithms like KNN. Standardization (z-score) centers data around mean → and standard deviation: $x' = \frac{x-\mu}{\sigma}$, ideal algorithms assuming Gaussian distributions such as linear regression or SVM.

- Encoding categorial data:- One-hot encoding transforms categories into binary vectors (e.g. "red", "blue" → 1), avoiding ordinal assumptions but increasing dimentionality. ~~table~~ Label encoding assigns integers (e.g. "low" = 0, "medium" = 1) suitable for tree-based models but risky for linear models due to implied order.

- Discretization and ~~Bi~~ Binning:- Discretization converts continous value into bins via equal-frequency or entropy-based methods, simplifying analysis and handling outliers.

- Dimentionality Reduction:- Principal Component Analysis (PCA) projects data onto fewer orthogonal components capturing maximum variance, combating the curse of dimentionality. other methods include t-SNE for visualisation or feature selection via statistical tests.

- Aggregation and generalization:- Aggregation summarizes data (e.g. daily sales to monthly totals using mean/sum), reducing volume while retaining trends. Generalization replaces low-level attributes with broader concepts (e.g. street to city).

- Other transformations:- Log transformations $\log(x+1)$ reduces right-skewness in positives like income. Smoothing (binning + averaging) removes noise. Feature Engineering creates new variables (e.g. interaction terms).

⇒ These techniques enhance model accuracy, reduce bias, and improve efficiency. e.g - scaled data boosts KNN performance by equalizing feature influence. In Python, sci-kit-learn provides tools like StandardScaler and OneHotEncoder for implementation. Proper transformation, ensures robust, interpretable results in data science, pipelines.